

H524 Introduction to Biostatistics
Week 5: Hypothesis Testing
Homework Assignment
Fall 2025

Due Date: End of Week 5

Total Points: 65 (+ 5 bonus)

Format: Submit through Canvas

Time Limit: 240 minutes (4 hours)

Instructions

- Show all work for full credit
- Round probabilities to 4 decimal places unless otherwise specified
- Include R code where specified
- You may use R to check your work

1 Part 1: Hypothesis Testing Fundamentals (18 points)

Question 1: Null Hypothesis Definition (3 points)

Which statement BEST describes the null hypothesis (H_0) in hypothesis testing?

- A) H_0 represents the research hypothesis; it is what we hope to prove through our study
- B) H_0 represents the alternative explanation; it is always what we want to reject
- C) H_0 is the hypothesis with the larger p-value; it cannot be proven, only disproven
- D) H_0 represents the status quo or no effect; it is the hypothesis we assume true until we have sufficient evidence to reject it

Question 2: Alternative Hypothesis Forms (3 points)

A researcher wants to test if a new drug LOWERS blood pressure compared to the current standard treatment (mean = 140 mmHg). Which alternative hypothesis is appropriate?

- A) $H_a: \mu > 140$ (one-sided, upper tail)
- B) $H_a: \mu \neq 140$ (two-sided)
- C) $H_a: \mu < 140$ (one-sided, lower tail)
- D) $H_0: \mu = 140$ (null hypothesis)

Question 3: P-value Interpretation (4 points)

Which statement CORRECTLY interprets what a p-value represents?

- A) The probability that the null hypothesis is true given our data
- B) The probability that the alternative hypothesis is false
- C) The probability of observing data as extreme as (or more extreme than) what we observed, assuming the null hypothesis is true
- D) The probability of making a Type I error in this specific test

Question 4: Decision Rule (2 points)

A hypothesis test is conducted at significance level $\alpha = 0.05$. The calculated p-value is 0.032. What is the correct decision?

- A) Fail to reject H_0 because p-value > 0.01 ; evidence is not strong enough
- B) Reject H_0 because p-value $< \alpha$; we have sufficient evidence against the null hypothesis
- C) Accept H_0 because p-value is close to α ; null hypothesis is proven true
- D) Reject H_a because p-value is small; alternative hypothesis is false

Question 5: T-statistic Calculation (3 points)

A researcher tests $H_0: \mu = 50$ vs $H_a: \mu \neq 50$. The sample statistics are: $n = 25$, $\bar{x} = 52$, $s = 8$. What is the t-statistic?

- A) 1.25
- B) 2.50
- C) 6.25
- D) 0.25

Question 6: P-value from t-statistic (3 points)

Given the scenario with ($t = 1.25$, $df = 24$, two-sided test), what is the p-value?

- A) 0.2234
- B) 0.0500
- C) 0.1117
- D) 0.8883

2 Part 2: Type I and Type II Errors (14 points)

Question 7: Type I Error (3 points)

What is a Type I error in hypothesis testing?

- A) Failing to reject the null hypothesis when it is actually false (false negative)
- B) Accepting the alternative hypothesis when it is false
- C) Rejecting the null hypothesis when it is actually true (false positive)
- D) Making a calculation error in computing the test statistic

Question 8: Type II Error (3 points)

A medical researcher tests a new drug that actually IS effective, but the hypothesis test fails to detect the effect. What type of error has occurred?

- A) Type I error (α); rejected a true null hypothesis
- B) Type II error (β); failed to reject a false null hypothesis
- C) Power error; the test was not powerful enough
- D) No error; the test correctly failed to reject H_0

Question 9: Error Trade-off (3 points)

What happens to the probability of Type II error (β) when you DECREASE the significance level (α) from 0.05 to 0.01?

- A) β decreases; being more conservative reduces all types of errors
- B) β stays the same; Type I and Type II errors are independent
- C) β becomes undefined; we can only control one type of error at a time
- D) β increases; making it harder to reject H_0 increases the chance of missing a true effect

Question 10: Statistical Power (2 points)

What is statistical power, and why is it important?

- A) Power = $1 - \alpha$; it's the probability of correctly accepting a true null hypothesis
- B) Power = $1 - \beta$; it's the probability of correctly rejecting a false null hypothesis (detecting a true effect)
- C) Power = $\alpha + \beta$; it measures the total probability of making any error
- D) Power = sample size / population size; larger samples have more power

Question 11: Blood Pressure Hypothesis Test (10 points)

A cardiologist wants to test if patients in a cardiac rehabilitation program have systolic blood pressure significantly **higher** than the normal threshold of 120 mmHg. She measures blood pressure for 20 patients:

```
bp_data <- c(118, 124, 130, 128, 122, 135, 119, 126, 132, 120,  
            125, 129, 121, 127, 123, 131, 120, 128, 124, 126)
```

Using R to perform a one-sample t-test:

- a) **(5 points)** What is the **t-statistic**? (Round to 2 decimal places)

$t = [t_stat]$

- b) **(5 points)** What is the **p-value**? (Round to 4 decimal places or use scientific notation)

p-value = [p-value]

Hint: Use `t.test(bp_data, mu = 120, alternative = "greater")`

3 Part 3: Hypothesis Testing with R (10 points)

Question 12: Cholesterol Hypothesis Test (10 points)

A pharmaceutical company claims their new statin drug reduces cholesterol to **below 200 mg/dL** on average. A clinical trial of 40 patients finds:

- Sample mean: $\bar{x} = 185$ mg/dL
- Sample standard deviation: $s = 35$ mg/dL

Test the company's claim using a one-sample t-test at the 0.01 significance level:

- a) **(5 points)** What is the **t-statistic**? (Round to 2 decimal places)

$t = [\text{t_stat}]$

- b) **(5 points)** What is the **p-value**? (Round to 4 decimal places)

$\text{p-value} = [\text{p_value}]$

Hint: The claim is "below 200", so $H_0: \mu = 200$ vs $H_a: \mu < 200$

4 Part 4: Two-Sample Tests (14 points)

Question 13: Two-Sample Test Application (3 points)

Which research question requires a TWO-SAMPLE t-test rather than a one-sample t-test?

- A) Testing if mean patient satisfaction score differs from the national average of 7.5
- B) Comparing mean recovery times between patients receiving Drug A versus Drug B
- C) Determining if mean BMI in a clinic exceeds 25
- D) Testing if mean blood glucose level equals 100 mg/dL

Question 14: Two-Sample T-test (4 points)

Two groups of patients are compared:

- Treatment A: $n_1 = 30$, $\bar{x}_1 = 145$, $s_1 = 15$
- Treatment B: $n_2 = 35$, $\bar{x}_2 = 138$, $s_2 = 18$

Testing $H_0: \mu_1 = \mu_2$ vs $H_a: \mu_1 \neq \mu_2$, which statement is correct?

- A) $t \approx 1.71$, p-value ≈ 0.0922 ; fail to reject H_0 at $\alpha = 0.05$
- B) $t \approx 3.21$, p-value ≈ 0.0018 ; fail to reject H_0 at $\alpha = 0.05$
- C) $t \approx 1.71$, p-value ≈ 0.0461 ; reject H_0 at $\alpha = 0.05$
- D) $t \approx 0.47$, p-value ≈ 0.6401 ; reject H_0 at $\alpha = 0.05$

Question 15: R Code for Two-Sample Test (2 points)

Given two vectors `group_a` and `group_b`, which R command correctly performs a two-sample t-test?

- A) `t.test(group_a - group_b)`
- B) `t.test(mean(group_a), mean(group_b))`
- C) `two.sample.test(group_a, group_b)`
- D) `t.test(group_a, group_b)`

Submission Requirements

Required Files:

1. **PDF Document** with all written work, calculations, and interpretations
2. **R Script File (.R)** with all R code properly commented (for Questions 11 and 12)

Formatting Guidelines:

- Clearly label each question number
- Show all calculation steps for full credit
- Round consistently per instructions
- Include units where appropriate (mmHg, mg/dL, etc.)

Grading Rubric

Category	Points	Criteria
Conceptual Understanding	24	Correct interpretation of hypotheses, errors, power
Calculations	20	Accurate computation of test statistics and p-values
R Implementation	11	Working code, proper syntax, correct results
Interpretation	10	Clear reasoning, appropriate conclusions

Common Mistakes to Avoid

- **Hypotheses:** Confusing H_0 and H_a ; using wrong tail (one-sided vs. two-sided)
- **P-values:** Misinterpreting p-value as probability that H_0 is true
- **Decision Rule:** Rejecting when p-value $> \alpha$, or failing to reject when p-value $< \alpha$
- **Errors:** Confusing Type I (α) and Type II (β) errors
- **R Code:** Using wrong `alternative` parameter (“greater”, “less”, “two.sided”)

Key Formulas

One-sample t-statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Two-sample t-statistic (equal variances):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Decision Rule:

- Reject H_0 if p-value $< \alpha$
- Fail to reject H_0 if p-value $\geq \alpha$

Relationships:

- Type I Error: $\Pr(\text{Reject } H_0 \mid H_0 \text{ true}) = \alpha$
- Type II Error: $\Pr(\text{Fail to reject } H_0 \mid H_0 \text{ false}) = \beta$
- Power: $1 - \beta = \Pr(\text{Reject } H_0 \mid H_0 \text{ false})$

Academic Integrity

All submitted work must be your own. While you may discuss problems with classmates, each student must write up their own solutions independently. Show all work and reasoning for full credit.

Resources

- **Textbook:** Pagano & Gauvreau, Chapter 9
- **Lecture Slides:** Week 5 slides with examples
- **Lab Materials:** Week 5 lab with step-by-step R code
- **R Documentation:** `?t.test`
- **Office Hours:** Wednesday 10:00-11:30am or by appointment